

Not Finished, But Abandoned

by

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ACKNOWLEDGMENTS

It is customary for authors of academic books to include in their prefaces statements such as this: "I am indebted to ... for their invaluable help; however, any errors which remain are my sole responsibility." Occasionally an author will go further. Rather than say that if there are any mistakes then he is responsible for them, he will say that there will inevitably be some mistakes and he is responsible for them....

Although the shouldering of all responsibility is usually a social ritual, the admission that errors exist is not — it is often a sincere avowal of belief. But this appears to present a living and everyday example of a situation which philosophers have commonly dismissed as absurd; that it is sometimes rational to hold logically incompatible beliefs.

— DAVID C. MAKINSON (1965)

Above is the famous “preface paradox,” which illustrates how to use the `wbepi` environment for epigraphs at the beginning of chapters. You probably also want to thank the Academy.

CONTENTS

Contents ii

List of Tables xviii

List of Figures xix

Abstract xxii

1 Introduction 1

1.1	<i>What the heck is a plasma</i>	1
1.1.1	Plasmas in the universe	1
1.1.2	How do plasmas affect us on earth?	1
1.1.3	Where does magnetic reconnection play into this whole shabang?	1
1.2	<i>Magnetic Reconnection</i>	1
1.2.1	The Generalized Ohm's Law	1
1.2.1.1	How do we derive this?	1
1.2.1.2	A breakdown of the terms	1
1.2.2	Regimes of Reconnection	1
1.2.2.1	Reconnection scaling parameters	1
1.2.2.2	Where do various systems line up?	2
1.2.2.3	What are we interested in?	2
1.2.3	Sweet-Parker Reconnection	2
1.2.3.1	A quick historical detour	2
1.2.3.2	What are the assumptions that go into Sweet-Parker?	2
1.2.3.3	Sweet-Parker Reconnection Rate Derivation	2
1.2.3.4	Application to Real Systems	2

1.2.3.4.1	What are some example parameters where reconnection occurs? .	2
1.2.3.4.2	What time scales does Sweet-Parker predict?	2
1.2.3.4.3	What did we miss?	2
1.2.4	Hall Reconnection	2
1.2.4.1	Why should the Hall term appear?	2
1.2.4.1.1	What is the relevant physics that sets when the hall term matters .	2
1.2.4.1.2	In what kind of plasmas does the hall term appear and when does it not?	2
1.2.4.2	What does hall reconnection look like . . .	3
1.2.4.3	What's missing?	3
1.2.5	Kinetic Reconnection	3
1.2.5.1	What makes kinetic reconnection "kinetic"? .	3
1.2.5.1.1	What do collisions do?	3
1.2.5.1.2	What makes a system kinetic? . .	3
1.2.5.2	Where does Pressure Anisotropy come from? .	3
1.2.5.2.1	Movement of flux tube	3
1.2.5.2.2	Effect of background field	3
1.2.5.3	How does pressure anisotropy change what the reconnection looks like	4
1.2.5.3.1	Thinner reconnection layer?	4
1.2.5.3.2	Marginal Firehose Condition . . .	4
1.2.5.3.3	Concluding: what features should we look for in the data?	4
1.2.5.4	Why do we want to measure it?	4
1.2.5.4.1	Simulation result overview	4
1.2.5.4.2	Spacecraft result overview	4

1.2.5.4.3	Experimental result?	4
1.3	<i>Experimental Setup</i> 5	
1.3.1	Overall how does our experiment progress	5
1.3.2	Initial field	6
1.3.2.1	Helmholtz	6
1.3.2.2	Permanent magnets	6
1.3.2.3	Toroidal	6
1.3.2.3.1	Shifted coil	6
1.3.2.3.2	Axial coil	6
1.3.2.4	Perturbation coils	6
1.3.3	Plasma Generation	6
1.3.3.1	How do our guns work	6
1.3.3.1.1	Gas lines	6
1.3.3.1.2	PFNs	6
1.3.3.1.3	Puff trigger	6
1.3.3.1.4	How to breakdown	6
1.3.3.2	How are the guns placed for injection? . .	6
1.3.3.2.1	How much do they fill the machine	6
1.3.3.2.2	What sets the filling fraction . . .	6
1.3.3.3	What goes wrong with them	7
1.3.3.3.1	Burning up	7
1.3.3.3.2	Burn holes in wall because of back- ground field	7
1.3.3.3.3	Air cooling	7
1.3.3.3.4	Optimizing gun operation	7
1.3.4	Drive field	7
1.3.4.1	Individual coils	7
1.3.4.1.1	Zone's between coils as plasma sources	7
1.3.4.2	Drive Cylinder	7
1.3.5	Diagnostics Overview	7

1.3.5.1	Bdot probes	7
1.3.5.1.1	Basic principles	7
1.3.5.1.2	Probe types	7
1.3.5.1.3	Data Interpretation	8
1.3.5.2	Hall probes	8
1.3.5.2.1	Basic principles	8
1.3.5.2.2	Issues and feasibility	8
1.3.5.3	Langmuir probes	8
1.3.5.3.1	Basic principles	8
1.3.5.3.2	What do we need to do to make them work for our device?	8
1.3.5.3.3	Look into XXX chapter	8
1.3.5.4	Fast camera	8
1.3.5.4.1	What can we see with it?	8
1.3.5.4.2	Potential use cases	8
1.3.6	How does the experiment compare to space	9
1.3.6.1	Simulation benefits and results	9
1.3.6.2	Space benefits and results	9
1.3.6.3	Lab benefits and results	9
1.3.7	Other lab reconnection experiments	9
1.3.7.1	VTF, MRX, FLARE	9
1.3.7.2	Exploding wire arrays	9
1.3.7.3	Laser based	9

2 Pressure Wave 10

2.1 Introduction 10

2.2 Observations that lead us to look at this wave 11

2.2.1	Speed probe measurements	11
2.2.1.1	Flux expansion before shock layer	11
2.2.1.2	Wavefront that depends on configuration	11
2.2.2	Simulations	11

2.2.2.1	Different n and drive change feature	11
2.2.2.2	Change reconnection layer post reflection .	11
2.3	<i>Physical understanding of wave</i>	11
2.3.1	Initial plasma configuration	11
2.3.2	Release in magnetic pressure	11
2.3.2.1	What would happen if plasma was immo- bile (replaced with perfect conductor) . .	11
2.3.3	Wavefront tells plasma about pressure Release . . .	11
2.3.3.1	Analogous system	11
2.3.4	Modeling magnetic field response	12
2.3.4.1	How do we create this model	12
2.3.4.2	Pressure wave is like a shield	12
2.3.4.3	Example	12
2.3.4.4	Why have we mentioned this?	12
2.3.4.4.1	Important for probe calibrations .	12
2.3.4.4.2	Good for predicting effects of dif- ferent coil configurations	12
2.4	<i>Wave Simulation</i>	12
2.4.1	Approximations	12
2.4.2	Linearizing MHD	12
2.4.3	Plasma Equilibrium + Symmetries	12
2.4.3.1	Types of Equilibrium	12
2.4.3.1.1	theta-pinch	12
2.4.3.1.2	screw-pinch	12
2.4.3.2	Types of Symmetries	12
2.4.3.2.1	z-symmetry	12
2.4.3.2.2	toroidal-symmetry	12
2.4.4	Simplified 1D equation	13
2.4.4.1	What does this look like?	13
2.4.4.2	Normalizations	13

2.4.4.3	Normalized equation	13
2.4.5	Boundary conditions	13
2.4.5.1	$r = 0$ Boundary	13
2.4.5.1.1	$v = 0$ option	13
2.4.5.1.2	$\xi = 0$ option	13
2.4.5.1.3	What do we choose?	13
2.4.5.2	$r = R$ Boundary	13
2.4.5.2.1	What is physically happening at high radius in the coils?	13
2.4.5.2.2	Loop voltage sets dB_z/dt	13
2.4.5.2.3	Assuming MHD so far so let's keep assuming!	13
2.4.5.2.4	Vloop signal shape	13
2.4.6	Initial conditions	14
2.4.6.1	What form can we expect plasma to take? .	14
2.4.6.2	What form do we use for these results . . .	14
2.4.7	What does our simulation show?	14
2.4.7.1	There is a wavefront traveling inwards at fast magnetosonic	14
2.4.7.1.1	Low beta approximation giving Alfven	14
2.4.7.2	Once wavefront passes, plasma moves out	14
2.4.7.3	Converting to B_{dot}	14
2.4.7.4	In B_{dot} we see flux moving outwards . . .	14
2.5	<i>How do we apply it to real data?</i>	14
2.5.1	Base requirements	14
2.5.1.1	Cylindrical symmetry	14
2.5.1.2	Releasing magnetic pressure at high radius	14
2.5.1.3	Measurements along a radial chord	14
2.5.1.4	Known initial field	14

2.5.2	Extracting the wave speed from the data	14
2.5.2.1	Wave tracking	14
2.5.2.1.1	Edge finding	14
2.5.2.1.2	Inertia	15
2.5.2.1.3	Monte-carlo	15
2.5.2.1.4	Centered finite difference	15
2.5.2.1.5	Mean + Standard Deviation of Speed	15
2.5.3	Getting the density	15
2.5.3.1	Using the alfven speed	15
2.5.4	Minimum measurable density	15
2.5.4.1	Assuming sample resolution	15
2.5.4.1.1	What does this assumption derive	15
2.5.4.1.2	How does it compare to the data .	15
2.5.4.1.3	We need a better approximation .	15
2.5.4.2	Assuming sub-sample resolution	16
2.5.4.3	A more complex model assuming the plasma is a step function * linear	16
2.6	<i>Does it reproduce langmuir probe results</i>	16
2.6.1	Experimental setup	16
2.6.1.1	Drive cylinder	16
2.6.1.2	Te probe along equator	16
2.6.2	Results	16
2.6.2.1	Langmuir probe minimum measurable den- sity	16
2.6.2.2	Impacts of simple Langmuir model vs BRL	16
2.6.2.3	What happens to the wave shape in simu- lation vs. experiment	16
2.6.2.3.1	Two-fluid dispersion	16
2.6.2.3.2	How might we expect this to change the wave	17

2.6.2.3.3	How to simulate evolution approx- imately	17
2.6.2.3.4	Dispersion simulation results . . .	17
2.7	<i>Applications</i>	17
2.7.1	Other machines where this is applicable	17
2.7.2	Diagnosing gun failures	18
2.7.3	Measuring effect of different gun number and Helmholtz strength	18
2.7.3.1	Gun number	18
2.7.3.2	Helmholtz	18
2.7.4	Measuring diffusion coefficient	18
2.7.4.1	Diffusion theoretical model	18
2.7.4.2	Diffusion simulations	18
2.7.4.3	Applications of pressure wave model . . .	18
2.8	<i>Future Research</i>	19
2.8.1	Better wave tracking	19
2.8.1.1	Cross-correlation	19
2.8.1.2	Dynamic Time Warping	19
2.8.1.3	Using a model of what the wave looks like	19
2.8.2	Using fast-magnetosonic speed instead of alfvén . .	19
2.8.2.1	Will this iteratively converge?	19
3	Multi-tip Langmuir Probe	20
3.1	<i>What's a Langmuir probe</i>	20
3.2	<i>Hardware</i>	20
3.2.1	Shaft Components	20
3.2.1.1	Garages	20
3.2.1.2	Shaft materials	20
3.2.1.3	Centering Collet	20
3.2.1.4	Feed-thru	20
3.2.2	Tip Components	20

3.2.2.1	Probe body	20
3.2.2.2	Tip material + size	20
3.2.2.2.1	Why use Mo	20
3.2.2.2.2	Why use the size we chose	20
3.2.2.3	Tip Shielding	20
3.2.2.3.1	Why shield at all?	20
3.2.2.3.2	First order mach effects to avoid	20
3.2.3	Bdot Coils	21
3.2.3.1	How many wraps	21
3.2.3.2	Approximate area	21
3.2.4	Length Scales	21
3.3	<i>Circuitry</i> 21	
3.3.1	Bias Source	21
3.3.1.1	Manually Controlled	21
3.3.1.1.1	Principles of operation	21
3.3.1.1.2	Problems	21
3.3.1.2	Remote Controlled	21
3.3.1.2.1	Principles of operation	21
3.3.1.2.2	Problems	21
3.3.1.2.3	Calibration	22
3.3.2	Probe Circuit	22
3.3.2.1	Principles of operation	22
3.3.2.1.1	Plasma draws current	22
3.3.2.1.2	Voltage drop across sense resistor	22
3.3.2.1.3	Measure high frequency voltage drop	22
3.3.2.2	Single-ended measurement	22
3.3.2.3	Differential measurement	22
3.3.2.4	Circuit Solution	22
3.3.2.4.1	How do we solve this	22

3.3.2.4.2	equations	23
3.3.2.4.3	Assumptions	23
3.3.2.5	Calibration	23
3.3.2.5.1	Calibrating in isat	23
3.3.2.5.2	Calibrating on the exponential . .	23
3.3.2.5.3	Bayesian calibration	23
3.3.2.5.4	Calibration using pressure wave .	23
3.3.3	Bdot coils	23
3.3.3.1	In-shaft circuit	23
3.3.3.1.1	Why do we have that?	23
3.3.3.1.2	Where is it placed approximately? .	23
3.3.3.2	Circuit model	24
3.3.3.3	Calibration	24
3.3.3.3.1	Directional	24
3.3.3.3.2	Transfer Function	24
3.4	<i>Plasma Theory</i> 24	
3.4.1	Simple Plasma Theory (Hutchinson)	24
3.4.1.1	Derivation	24
3.4.1.2	Pros & Cons	24
3.4.2	BRL	24
3.4.2.1	Idea behind it	24
3.4.2.2	Approximations of it	24
3.4.2.2.1	Planar probe	24
3.4.2.2.2	Chen Approximation	24
3.4.2.2.3	Even more approximation	24
3.4.3	Effect of electric fields	24
3.4.3.1	Where does the electric field come from . .	24
3.4.3.1.1	Majority inductive	24
3.4.3.1.2	Some static	25
3.4.3.2	What does it do to the tip measurements .	25

3.4.3.2.1	Theoretical understanding	25
3.4.3.2.2	Physical reality	25
3.4.4	Minimum measurable density	26
3.4.4.1	What's the debye Length	26
3.4.4.2	Sheath scale	26
3.4.4.3	Tip-tip sheath overlap	26
3.5	<i>Curve Fitting</i> 26	
3.5.1	Least squares Fitting	26
3.5.1.1	Benefits (easy)	26
3.5.1.2	Issues (not representative of error)	26
3.5.2	ODR	26
3.5.2.1	What does error look like in our signals . .	26
3.5.2.2	How does ODR take that into account . . .	26
3.5.2.3	How to apply parameter bounds	26
3.5.2.3.1	Our method	26
3.5.2.3.2	Other options	26
3.5.3	Monte-carlo Fitting	27
3.5.3.1	What kind of fits do we get if we just trust a local minimum	27
3.5.3.2	How do we sample initial conditions . . .	27
3.5.3.3	More robust methods for getting global minimums	27
3.5.4	Combining multiple shots	27
3.5.4.1	Using some time points from each shot . .	27
3.5.4.2	Combining all IV points and fitting that . .	27
3.5.4.3	Problems	27
3.5.4.3.1	Timing alignment	27
3.5.4.3.2	Variability of plasma parameters . .	27
3.6	<i>Results</i> 27	
3.6.1	Measurement of low densities	28

3.6.1.1	How does Hutchinson vs BRL treat ion saturation	28
3.6.1.2	What are the experimental predictions that Hutchinson vs BRL have	28
3.6.1.3	How does that change the results	28
3.6.2	Noise resilience	28
3.6.3	Fits more accurate	28
3.6.4	Extracting electric fields	28
3.6.4.1	Where do we expect electric fields to be . .	28
3.6.4.2	Do we see them there	28
4	Pressure Anisotropy Probe 29	
4.1	<i>How can we measure Pressure Anisotropy</i> 29	
4.1.1	Thomson Scattering	29
4.1.1.1	Pros	29
4.1.1.2	Cons	29
4.1.2	Directional Langmuir Probe	29
4.1.2.1	Pros	29
4.1.2.2	Cons	29
4.2	<i>Hardware</i> 30	
4.2.1	Shaft Components	30
4.2.1.1	Garages	30
4.2.1.2	Shaft materials	30
4.2.1.3	Centering Collet	30
4.2.1.4	Feed-thru	30
4.2.2	Tip Components	30
4.2.2.1	Probe body	30
4.2.2.1.1	3D printed	30
4.2.2.1.2	Both sides fit together	30
4.2.2.2	Tip material + size	30
4.2.2.2.1	Shell material	30

4.2.2.2.2	Core material	30
4.2.2.3	Tip Shielding	31
4.2.2.3.1	Shell Shielding	31
4.2.2.3.2	Core Shielding	31
4.2.3	Bdot Coils	31
4.2.4	Connecting Bdot - Probe - Shaft	31
4.2.5	Length Scales	31
4.3	<i>Circuitry</i> 31	
4.3.1	Bias Source	31
4.3.1.1	Principles of operation	31
4.3.1.1.1	Charging	31
4.3.1.1.2	Discharging	31
4.3.1.1.3	Swapping polarity	31
4.3.1.2	How do we do this	31
4.3.1.2.1	Cannibalization	31
4.3.1.2.2	Discharging	32
4.3.1.2.3	Safety	32
4.3.1.2.4	Cap arcing	32
4.3.1.2.5	Reducing voltage drop due to inductance	32
4.3.1.2.6	Reversing polarity	32
4.3.1.2.7	LabView (ew)	32
4.3.1.3	Calibration	33
4.3.1.3.1	Calibrating power supply output	33
4.3.1.3.2	Calibrating cap bank output	33
4.3.2	Probe Circuit	33
4.3.2.1	Single-ended measurement	33
4.3.2.1.1	TREX cells	33
4.3.2.1.2	Why was this bad	33
4.3.2.1.3	How long did we spend on this :(.	33

4.3.2.2	Differential measurement	33
4.3.2.2.1	Parallel circuitry	33
4.3.2.3	Figuring out grounds	33
4.3.2.3.1	Types of grounds on the ACQ . .	33
4.3.2.3.2	Routing grounds	33
4.3.2.4	Swapping current paths	34
4.3.2.4.1	Why do we want to do this	34
4.3.2.4.2	Circuitry to do this	34
4.3.2.4.3	How does this change results . . .	34
4.3.2.5	Circuit Solution	34
4.3.2.5.1	Solution	34
4.3.2.5.2	Assumptions	34
4.3.2.6	Calibration	34
4.3.2.6.1	Calibrating in isat	34
4.3.2.6.2	Calibrating on the exponential . .	35
4.3.2.6.3	Rotation calibration	35
4.3.2.7	The path to ground	35
4.3.2.7.1	Why do we have to care	35
4.3.2.7.2	How do we set it	35
4.3.2.8	Capacitive coupling	35
4.3.2.8.1	Where does it come from	35
4.3.2.8.2	Where do we see it	35
4.3.2.8.3	Magnitude of effect	35
4.3.2.8.4	Removing effect	35
4.4	<i>Plasma Theory</i> 36	
4.4.1	A full model	36
4.4.1.1	Plasma Distribution	36
4.4.1.2	Ion particle flux	36
4.4.1.2.1	Mach effects	36
4.4.1.3	Electron particle flux	36

4.4.1.3.1	Minimum velocity	36
4.4.1.3.2	Area	36
4.4.1.3.3	Particle Flux	36
4.4.1.4	Theoretical JV curves	36
4.4.1.4.1	Reading radar plots	36
4.4.2	Individual tip fitting	36
4.4.2.1	Why do this instead	36
4.4.2.2	Methods	36
4.4.2.2.1	Aforementioned methods	36
4.4.2.2.2	Exponential	37
4.4.2.3	Extracting plasma parameters	37
4.4.2.3.1	Spherical harmonics	37
4.4.2.3.2	Error bars	37
4.5	<i>Curve Fitting</i> 37	
4.5.1	Combining multiple shots	38
4.5.1.1	Timing alignment	38
4.5.1.1.1	$B_z = 0$ alignment	38
4.5.1.1.2	B_{dot} cross correlation	38
4.5.1.1.3	Result comparison	38
4.5.1.2	Averaging of core data	38
4.5.1.2.1	Weighted average of relay on/off	38
4.5.2	Fitting shell	39
4.5.2.1	Use all shell data	39
4.5.2.2	Fit using BRL	39
4.5.2.3	Fits pretty good	39
4.5.3	Fitting core	39
4.5.3.1	Fitting using full routine	39
4.5.3.2	Fitting tips independently	39
4.6	<i>Results</i> 39	
4.6.1	Drive cylinder	39

4.6.1.1	What did we try	39
4.6.1.2	Why didn't we see pressure anisotropy . .	39
4.6.1.2.1	In the bdot data	39
4.6.1.2.2	In the PA data	39
4.6.2	3 coil	39
4.6.2.1	How much anisotropy should we expect .	39
4.6.2.2	What setups did we try	39
4.6.2.3	Old results of just comparing radar plots .	39
4.6.2.4	New results using independent fitting . . .	39
5	Conclusion	40
5.1	<i>Summary of thesis</i>	40
5.1.1	Using pressure waves to measure density	40
5.1.2	Upgrading langmuir probe analysis and new possibilities	40
5.1.3	Using directional langmuir probes for measuring pressure anisotropy	40
5.2	<i>Future work</i>	40
5.2.1	Pressure wave	40
5.2.1.1	Aforementioned future work	40
5.2.1.2	Extracting cross-field diffusion	40
5.2.2	Langmuir Probe	40
5.2.2.1	Prepared to analyze lots of cool data	40
5.2.2.2	Extract out electric field	40
5.2.3	PA Probe	40
5.2.3.1	Try more configurations	40
5.2.3.2	Map out full pressure anisotropy during shot	40
	Colophon	41

LIST OF TABLES

1.1	List of diagnostics used in the experiment and what they can measure.	7
1.2	Lab vs. Space Parameters	9
2.1	Parameters for normalization	13
3.1	Length scales of probe	21
3.2	Bias circuit component values	22
3.3	Probe circuit component values	23
3.4	Chen parameters	24
4.1	Length scales of probe	31
4.2	Bias circuit component values	33
4.3	Probe circuit component values	33

LIST OF FIGURES

1.1	Cartoon of places where reconnection can happen.	1
1.2	Diagram of reconnection regimes.	1
1.3	Diagram of Sweet-Parker reconnection.	2
1.4	Diagram of 2d hall reconnection	3
1.5	Flux tube changing size during reconnection causing pressure anisotropy build up.	3
1.6	Simulated kinetic reconnection with changing guide field. . .	4
1.7	Diagram from Ohia of with and without pressure anisotropy in simulation.	4
1.8	Some figure from Jan's papers thowing effects of pressure anisotropy	4
1.9	Timeline of experimental sequence.	5
1.10	Panels from flux arrays of experiment happening.	5
1.11	Circuit diagram of plasma gun.	6
1.12	Simplified cut of plasma gun insides.	6
1.13	Image from high speed camera of gun spewing	7
2.1	dB_z/dt plots for different Helmholtz fields	10
2.2	Greess et al. 2022 paper showing simulations with pressure wave.	10
2.3	Simple diagram of sound wave in container	12
2.4	Data vs. crude minimum density vs. empirical minimum density	15
2.5	Diagram of what the more complex model looks like.	15
2.6	CAD rendering of BRB w/ drive cylinder, flux probes, and Te probe.	16
2.7	Plot of langmuir density and wave density	16
2.8	Diagram of sheath overlap for multitip langmuir probe	16
2.9	Plot of two-fluid dispresion relation	16

2.10	Results of simulating application of dispresion relation to pres- sure wave	17
2.11	Initial density before and after many shots	17
2.12	Initial density for varying gun number	17
2.13	Initial density for varying Helmholtz	17
2.14	Diagram of flux mechanisms.	18
2.15	Example diffusion simulation	18
2.16	Example of cross-correlation	18
2.17	Example of DTW	19
3.1	Diagram of simple langmuir probe.	20
3.2	Figure 2.6 from Joe's thesis	20
3.3	Picture of end of probe	21
3.4	Probe body from engineering sketch	21
3.5	Circuit diagram of old setup. Add switches.	21
3.6	Image of old setup	22
3.7	Circuit diagram of new setup. Add switches.	22
3.8	Image of new setup	22
3.9	Simple circuit diagram	23
3.10	Image of old circuit board	23
3.11	Image of new circuit board	23
3.12	What does an electric field do to the tip measurements diagram.	25
3.13	Example plot where E splitting tips	25
3.14	Diagram of local minimums	26
3.15	Example fit using many shots, multiple local parameter fits, and errors	27
3.16	Plot of isat currents from BRL	27
3.17	Radial chord of initial density with Hutchinson vs BRL.	28
3.18	Comparison of expected E from Ohm's law and measured E.	28
4.1	Picture of feed thru along with drawings of cross-sections.	30

4.2	Picture of end of probe	30
4.3	CAD diagram of full body	30
4.4	CAD diagram of probe tip	30
4.5	Simple circuit diagram of how bias works.	31
4.6	Full circuit diagram of how bias works.	32
4.7	Image of power supply setup	32
4.8	Schematic of grounding diagram	33
4.9	Image of swapping box	34
4.10	Diagram of swapping	34
4.11	Time frame for core movie of IV with separation between relay	34
4.12	Circuit diagram of probe circuitry	35
4.13	Image of probe circuitry	35
4.14	Probe signals showing coupling	35
4.15	Figure of theoretial plasma distributions and JV curves	36
4.16	Simple drawing of what might happen to ions	37
4.17	Comparison of alignment using $B_z = 0$ or $\mathbf{b} \cdot \dot{\mathbf{r}}$ cross correlation	38

ABSTRACT

FIXME: basically a placeholder; do not believe

I did some research, read a bunch of papers, published a couple myself, (pick one):

1. ran some experiments and made some graphs,
2. proved some theorems

and now I have a job. I've assembled this document in the last couple of months so you will let me leave. Thanks!

1 INTRODUCTION

1.1 What the heck is a plasma

1.1.1 Plasmas in the universe

1.1.2 How do plasmas affect us on earth?

1.1.3 Where does magnetic reconnection play into this whole shabang?

1.2 Magnetic Reconnection

1.2.1 The Generalized Ohm's Law

1.2.1.1 How do we derive this?

1.2.1.2 A breakdown of the terms

1.2.2 Regimes of Reconnection

1.2.2.1 Reconnection scaling parameters

i.e. why do we use S and L/d_i to characterize reconnection regimes?

Figure 1.1: Cartoon of places where reconnection can happen.

Figure 1.2: Diagram of reconnection regimes.

Figure 1.3: Diagram of Sweet-Parker reconnection.

1.2.2.2 Where do various systems line up?

1.2.2.3 What are we interested in?

1.2.3 Sweet-Parker Reconnection

1.2.3.1 A quick historical detour

1.2.3.2 What are the assumptions that go into Sweet-Parker?

1.2.3.3 Sweet-Parker Reconnection Rate Derivation

1.2.3.4 Application to Real Systems

1.2.3.4.1 What are some example parameters where reconnection occurs?

1.2.3.4.2 What time scales does Sweet-Parker predict?

1.2.3.4.3 What did we miss?

1.2.4 Hall Reconnection

1.2.4.1 Why should the Hall term appear?

1.2.4.1.1 What is the relevant physics that sets when the hall term matters

1.2.4.1.2 In what kind of plasmas does the hall term appear and when does it not?

Figure 1.4: Diagram of 2d hall reconnection

Figure 1.5: Flux tube changing size during reconnection causing pressure anisotropy build up.

1.2.4.2 What does hall reconnection look like

1.2.4.3 What's missing?

1.2.5 Kinetic Reconnection

1.2.5.1 What makes kinetic reconnection “kinetic”?

1.2.5.1.1 What do collisions do?

1.2.5.1.1.1 How quickly do collisions damp out non-Maxwellian features?

1.2.5.1.2 What makes a system kinetic?

1.2.5.1.2.1 What parameters set this?

1.2.5.1.2.2 What should we do in the lab to get this?

1.2.5.2 Where does Pressure Anisotropy come from?

1.2.5.2.1 Movement of flux tube

1.2.5.2.2 Effect of background field

1.2.5.2.2.1 Too low of background field does...

1.2.5.2.2.2 Too high of field does...

Figure 1.6: Simulated kinetic reconnection with changing guide field.

Figure 1.7: Diagram from Ohia of with and without pressure anisotropy in simulation.

Figure 1.8: Some figure from Jan's papers showing effects of pressure anisotropy

1.2.5.2.2.3 Goldilocks!

1.2.5.3 How does pressure anisotropy change what the reconnection looks like

1.2.5.3.1 Thinner reconnection layer? Use results from Jan's more recent papers.

1.2.5.3.2 Marginal Firehose Condition

1.2.5.3.3 Concluding: what features should we look for in the data?

1.2.5.4 Why do we want to measure it?

At the end of experimental setup I'll talk about why we want to do it in the lab but I think motivating why we care about kinetic reconnection as a process would be good here.

1.2.5.4.1 Simulation result overview

1.2.5.4.2 Spacecraft result overview

1.2.5.4.3 Experimental result?

1.3 Experimental Setup

1.3.1 Overall how does our experiment progress

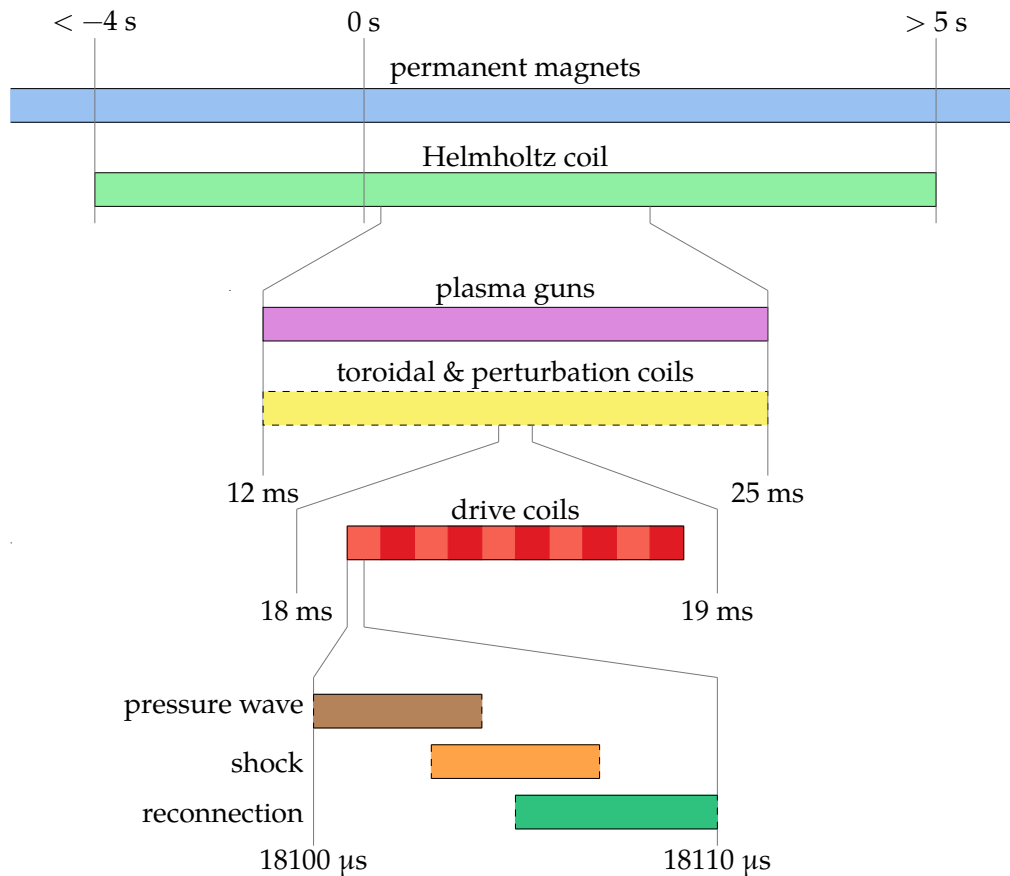


Figure 1.9: Timeline of experimental sequence.

Figure 1.10: Panels from flux arrays of experiment happening.

Figure 1.11: Circuit diagram of plasma gun.

Figure 1.12: Simplified cut of plasma gun insides.

1.3.2 Initial field

1.3.2.1 Helmholtz

1.3.2.2 Permanent magnets

1.3.2.3 Toroidal

1.3.2.3.1 Shifted coil

1.3.2.3.2 Axial coil

1.3.2.4 Perturbation coils

1.3.3 Plasma Generation

1.3.3.1 How do our guns work

1.3.3.1.1 Gas lines

1.3.3.1.2 PFNs

1.3.3.1.3 Puff trigger

1.3.3.1.4 How to breakdown

1.3.3.2 How are the guns placed for injection?

1.3.3.2.1 How much do they fill the machine

1.3.3.2.2 What sets the filling fraction

Figure 1.13: Image from high speed camera of gun spewing

Table 1.1: List of diagnostics used in the experiment and what they can measure.

1.3.3.3 What goes wrong with them

1.3.3.3.1 Burning up

1.3.3.3.2 Burn holes in wall because of background field Get image from Paul with hole in wall.

1.3.3.3.3 Air cooling

1.3.3.3.4 Optimizing gun operation

1.3.4 Drive field

1.3.4.1 Individual coils

1.3.4.1.1 Zone's between coils as plasma sources

1.3.4.2 Drive Cylinder

Reference Paul's RSI + thesis.

1.3.5 Diagnostics Overview

1.3.5.1 Bdot probes

1.3.5.1.1 Basic principles

1.3.5.1.2 Probe types

1.3.5.1.2.1 Linear probe

1.3.5.1.2.2 Hook probe

1.3.5.1.2.3 Speed probe

1.3.5.1.2.4 Flux array 1

1.3.5.1.2.5 Flux array 2

1.3.5.1.3 Data Interpretation How do we convert the raw data into something useful?

1.3.5.2 Hall probes

1.3.5.2.1 Basic principles

1.3.5.2.2 Issues and feasibility

1.3.5.3 Langmuir probes

1.3.5.3.1 Basic principles

1.3.5.3.2 What do we need to do to make them work for our device?

1.3.5.3.3 Look into XXX chapter

1.3.5.4 Fast camera

1.3.5.4.1 What can we see with it?

1.3.5.4.2 Potential use cases

Table 1.2: Comparison of lab and space parameters.

1.3.6 How does the experiment compare to space

1.3.6.1 Simulation benefits and results

1.3.6.2 Space benefits and results

1.3.6.3 Lab benefits and results

1.3.7 Other lab reconnection experiments

1.3.7.1 VTF, MRX, FLARE

1.3.7.2 Exploding wire arrays

1.3.7.3 Laser based

2 PRESSURE WAVE

2.1 Introduction

Once I get Jan's rewrite of the intro + conclusion I can figure out what might go here.

Figure 2.1: dBz/dt plots for different Helmholtz fields

Figure 2.2: Greess et al. 2022 paper showing simulations with pressure wave.

2.2 Observations that lead us to look at this wave

2.2.1 Speed probe measurements

2.2.1.1 Flux expansion before shock layer

2.2.1.2 Wavefront that depends on configuration

2.2.2 Simulations

2.2.2.1 Different n and drive change feature

2.2.2.2 Change reconnection layer post reflection

2.3 Physical understanding of wave

2.3.1 Initial plasma configuration

2.3.2 Release in magnetic pressure

2.3.2.1 What would happen if plasma was immobile (replaced with perfect conductor)

All field is shielded at edge and there is no change in magnetic pressure in the bulk.

2.3.3 Wavefront tells plasma about pressure Release

2.3.3.1 Analogous system

Sound wave.

How does a sound wave propagate in a plasma? Fast magnetosonic wave!

Figure 2.3: Simple diagram of sound wave in container

2.3.4 Modeling magnetic field response

2.3.4.1 How do we create this model

2.3.4.2 Pressure wave is like a shield

2.3.4.3 Example

2.3.4.4 Why have we mentioned this?

2.3.4.4.1 Important for probe calibrations

2.3.4.4.2 Good for predicting effects of different coil configurations

2.4 Wave Simulation

2.4.1 Approximations

2.4.2 Linearizing MHD

2.4.3 Plasma Equilibrium + Symmetries

2.4.3.1 Types of Equilibrium

2.4.3.1.1 theta-pinch

2.4.3.1.2 screw-pinch

2.4.3.2 Types of Symmetries

2.4.3.2.1 z-symmetry

2.4.3.2.2 toroidal-symmetry

Table 2.1: Parameters for normalization

2.4.4 Simplified 1D equation

2.4.4.1 What does this look like?

A wave equation!

2.4.4.2 Normalizations

2.4.4.3 Normalized equation

2.4.5 Boundary conditions

2.4.5.1 $r = 0$ Boundary

2.4.5.1.1 $v = 0$ option

2.4.5.1.2 $\xi = 0$ option

2.4.5.1.3 What do we choose?

2.4.5.2 $r = R$ Boundary

2.4.5.2.1 What is physically happening at high radius in the coils?

2.4.5.2.2 Loop voltage sets dB_z/dt

2.4.5.2.3 Assuming MHD so far so let's keep assuming!

2.4.5.2.4 Vloop signal shape

2.4.6 Initial conditions

2.4.6.1 What form can we expect plasma to take?

2.4.6.2 What form do we use for these results

2.4.7 What does our simulation show?

2.4.7.1 There is a wavefront traveling inwards at fast magnetosonic

2.4.7.1.1 Low beta approximation giving Alfven

2.4.7.2 Once wavefront passes, plasma moves out

2.4.7.3 Converting to B_{dot}

2.4.7.4 In B_{dot} we see flux moving outwards

As expected from MHD assumption.

2.5 How do we apply it to real data?

2.5.1 Base requirements

2.5.1.1 Cylindrical symmetry

2.5.1.2 Releasing magnetic pressure at high radius

2.5.1.3 Measurements along a radial chord

2.5.1.4 Known initial field

2.5.2 Extracting the wave speed from the data

2.5.2.1 Wave tracking

2.5.2.1.1 Edge finding

Figure 2.4: Data vs. crude minimum density vs. empirical minimum density

Figure 2.5: Diagram of what the more complex model looks like.

2.5.2.1.2 Inertia

2.5.2.1.3 Monte-carlo

2.5.2.1.4 Centered finite difference

2.5.2.1.5 Mean + Standard Deviation of Speed

2.5.3 Getting the density

2.5.3.1 Using the alfven speed

2.5.4 Minimum measurable density

2.5.4.1 Assuming sample resolution

2.5.4.1.1 What does this assumption derive

2.5.4.1.2 How does it compare to the data

2.5.4.1.3 We need a better approximation

Figure 2.6: CAD rendering of BRB w/ drive cylinder, flux probes, and Te probe.

Figure 2.7: Plot of langmuir density and wave density

Figure 2.8: Diagram of sheath overlap for multitip langmuir probe

2.5.4.2 Assuming sub-sample resolution

**2.5.4.3 A more complex model assuming the plasma is a step function
* linear**

2.6 Does it reproduce langmuir probe results

2.6.1 Experimental setup

2.6.1.1 Drive cylinder

2.6.1.2 Te probe along equator

2.6.2 Results

2.6.2.1 Langmuir probe minimum measurable density

2.6.2.2 Impacts of simple Langmuir model vs BRL

Refer to figure in Te section showing how it matters.

2.6.2.3 What happens to the wave shape in simulation vs. experiment

2.6.2.3.1 Two-fluid dispersion

Figure 2.9: Plot of two-fluid dispresion relation

Figure 2.10: Results of simulating application of dispersion relation to pressure wave

Figure 2.11: Initial density before and after many shots

Figure 2.12: Initial density for varying gun number

2.6.2.3.2 How might we expect this to change the wave

2.6.2.3.3 How to simulate evolution approximately

2.6.2.3.4 Dispersion simulation results

2.7 Applications

2.7.1 Other machines where this is applicable

LAPD & MRX

Figure 2.13: Initial density for varying Helmholtz

Figure 2.14: Diagram of flux mechanisms.

Figure 2.15: Example diffusion simulation

Figure 2.16: Example of cross-correlation

2.7.2 Diagnosing gun failures

2.7.3 Measuring effect of different gun number and Helmholtz strength

2.7.3.1 Gun number

2.7.3.2 Helmholtz

2.7.4 Measuring diffusion coefficient

2.7.4.1 Diffusion theoretical model

2.7.4.2 Diffusion simulations

2.7.4.3 Applications of pressure wave model

I think I need to do a bit more research here to show that we can actually extract a diffusion coefficient.

Figure 2.17: Example of DTW

2.8 Future Research

2.8.1 Better wave tracking

2.8.1.1 Cross-correlation

2.8.1.2 Dynamic Time Warping

2.8.1.3 Using a model of what the wave looks like

2.8.2 Using fast-magnetosonic speed instead of alfven

2.8.2.1 Will this iteratively converge?

3 MULTI-TIP LANGMUIR PROBE

3.1 What's a Langmuir probe

3.2 Hardware

3.2.1 Shaft Components

3.2.1.1 Garages

3.2.1.2 Shaft materials

3.2.1.3 Centering Collet

3.2.1.4 Feed-thru

3.2.2 Tip Components

3.2.2.1 Probe body

3.2.2.2 Tip material + size

3.2.2.2.1 Why use Mo

3.2.2.2.2 Why use the size we chose

3.2.2.3 Tip Shielding

3.2.2.3.1 Why shield at all?

3.2.2.3.2 First order mach effects to avoid

Figure 3.1: Diagram of simple langmuir probe.

Figure 3.2: Figure 2.6 from Joe's thesis

Figure 3.3: Picture of end of probe

Figure 3.4: Probe body from engineering sketch

Table 3.1: Length scales of probe

3.2.3 Bdot Coils

3.2.3.1 How many wraps

3.2.3.2 Approximate area

3.2.4 Length Scales

3.3 Circuitry

3.3.1 Bias Source

3.3.1.1 Manually Controlled

3.3.1.1.1 Principles of operation - Equation to use for bias calculation

3.3.1.1.2 Problems - Had to be manually controlled for drastic bias changes - Wires often contacted each other or parts would lose contact

3.3.1.2 Remote Controlled

3.3.1.2.1 Principles of operation - Equation to use for calculation

3.3.1.2.2 Problems - Destroyed a power supply - A resistor was designed and thus burnt out

Figure 3.5: Circuit diagram of old setup. Add switches.

Figure 3.6: Image of old setup

Figure 3.7: Circuit diagram of new setup. Add switches.

Figure 3.8: Image of new setup

3.3.1.2.3 Calibration

3.3.2 Probe Circuit

3.3.2.1 Principles of operation

3.3.2.1.1 Plasma draws current

3.3.2.1.2 Voltage drop across sense resistor

3.3.2.1.3 Measure high frequency voltage drop

3.3.2.2 Single-ended measurement

3.3.2.3 Differential measurement

3.3.2.4 Circuit Solution

3.3.2.4.1 How do we solve this

3.3.2.4.1.1 Kirkoff's laws

3.3.2.4.1.2 System of equations

3.3.2.4.1.3 SymPy

Table 3.2: Bias circuit component values

Figure 3.9: Simple circuit diagram

Figure 3.10: Image of old circuit board

Figure 3.11: Image of new circuit board

3.3.2.4.2 equations

3.3.2.4.3 Assumptions

3.3.2.5 Calibration

3.3.2.5.1 Calibrating in isat

3.3.2.5.2 Calibrating on the exponential

3.3.2.5.3 Bayesian calibration

3.3.2.5.4 Calibration using pressure wave

3.3.3 Bdot coils

3.3.3.1 In-shaft circuit

3.3.3.1.1 Why do we have that?

3.3.3.1.2 Where is it placed approximately?

Table 3.3: Probe circuit component values

Table 3.4: Chen parameters

3.3.3.2 Circuit model**3.3.3.3 Calibration****3.3.3.3.1 Directional****3.3.3.3.2 Transfer Function****3.4 Plasma Theory****3.4.1 Simple Plasma Theory (Hutchinson)****3.4.1.1 Derivation****3.4.1.2 Pros & Cons****3.4.2 BRL****3.4.2.1 Idea behind it****3.4.2.2 Approximations of it****3.4.2.2.1 Planar probe****3.4.2.2.2 Chen Approximation****3.4.2.2.3 Even more approximation****3.4.3 Effect of electric fields****3.4.3.1 Where does the electric field come from****3.4.3.1.1 Majority inductive**

Figure 3.12: What does an electric field do to the tip measurements diagram.

Figure 3.13: Example plot where E splitting tips

3.4.3.1.2 Some static

3.4.3.1.2.1 What is the expected static magnitude

3.4.3.2 What does it do to the tip measurements

3.4.3.2.1 Theoretical understanding

3.4.3.2.2 Physical reality

Figure 3.14: Diagram of local minimums

3.4.4 Minimum measurable density

3.4.4.1 What's the debye Length

3.4.4.2 Sheath scale

3.4.4.3 Tip-tip sheath overlap

3.5 Curve Fitting

3.5.1 Least squares Fitting

3.5.1.1 Benefits (easy)

3.5.1.2 Issues (not representative of error)

3.5.2 ODR

3.5.2.1 What does error look like in our signals

3.5.2.2 How does ODR take that into account

3.5.2.3 How to apply parameter bounds

3.5.2.3.1 Our method

3.5.2.3.2 Other options

Figure 3.15: Example fit using many shots, multiple local parameter fits, and errors

Figure 3.16: Plot of isat currents from BRL

3.5.3 Monte-carlo Fitting

3.5.3.1 What kind of fits do we get if we just trust a local minimum

3.5.3.2 How do we sample initial conditions

3.5.3.3 More robust methods for getting global minimums

3.5.4 Combining multiple shots

3.5.4.1 Using some time points from each shot

3.5.4.2 Combining all IV points and fitting that

3.5.4.3 Problems

3.5.4.3.1 Timing alignment We'll go into this more in the PA section.

3.5.4.3.2 Variability of plasma parameters

3.6 Results

What is the benefit of doing all this work?

Figure 3.17: Radial chord of initial density with Hutchinson vs BRL.

Figure 3.18: Comparison of expected E from Ohm's law and measured E.

3.6.1 Measurement of low densities

3.6.1.1 How does Hutchinson vs BRL treat ion saturation

3.6.1.2 What are the experimental predictions that Hutchinson vs BRL have

3.6.1.3 How does that change the results

3.6.2 Noise resilience

Not sure how best I should show this.

3.6.3 Fits more accurate

- Each fit uses many data points
- Don't have to arbitrarily cutoff data points

3.6.4 Extracting electric fields

3.6.4.1 Where do we expect electric fields to be

3.6.4.2 Do we see them there

4 PRESSURE ANISOTROPY PROBE

4.1 How can we measure Pressure Anisotropy

4.1.1 Thomson Scattering

4.1.1.1 Pros

4.1.1.2 Cons

- Requires a lot of playing around with optics
- Size of machine
- Don't have much in-lab expertise
- Getting all the photons is tough
- Don't have optics stuff setup

4.1.2 Directional Langmuir Probe

4.1.2.1 Pros

- Simplicity
- Cost
- Mobility

4.1.2.2 Cons

- Difficult to interpret results precisely (luckily only need relative stuff)
- Can't perfectly model
- Requires a lot of playing around with circuits

Figure 4.1: Picture of feed thru along with drawings of cross-sections.

Figure 4.2: Picture of end of probe

Figure 4.3: CAD diagram of full body

4.2 Hardware

4.2.1 Shaft Components

4.2.1.1 Garages

4.2.1.2 Shaft materials

4.2.1.3 Centering Collet

4.2.1.4 Feed-thru

4.2.2 Tip Components

4.2.2.1 Probe body

4.2.2.1.1 3D printed

4.2.2.1.2 Both sides fit together

4.2.2.2 Tip material + size

4.2.2.2.1 Shell material

4.2.2.2.2 Core material

Figure 4.4: CAD diagram of probe tip

Table 4.1: Length scales of probe

Figure 4.5: Simple circuit diagram of how bias works.

4.2.2.3 Tip Shielding

4.2.2.3.1 Shell Shielding

4.2.2.3.2 Core Shielding

4.2.3 Bdot Coils

Same as Te probe

4.2.4 Connecting Bdot - Probe - Shaft

4.2.5 Length Scales

4.3 Circuitry

4.3.1 Bias Source

4.3.1.1 Principles of operation

4.3.1.1.1 Charging

4.3.1.1.2 Discharging

4.3.1.1.3 Swapping polarity

4.3.1.2 How do we do this

4.3.1.2.1 Cannibalization

Figure 4.6: Full circuit diagram of how bias works.

Figure 4.7: Image of power supply setup

4.3.1.2.2 Discharging

4.3.1.2.3 Safety Diodes for discharging the wrong way.

4.3.1.2.4 Cap arcing

4.3.1.2.4.1 What happened

4.3.1.2.4.2 Solution

4.3.1.2.5 Reducing voltage drop due to inductance

4.3.1.2.5.1 Twisting Wires

4.3.1.2.5.2 Adding cap next to board

4.3.1.2.6 Reversing polarity

4.3.1.2.6.1 Why we break relay

4.3.1.2.6.2 How do we fix this problem

4.3.1.2.7 LabView (ew)

4.3.1.2.7.1 Controls CRIO

4.3.1.2.7.2 Countdown if too high of voltage for swapping

Table 4.2: Bias circuit component values

Table 4.3: Probe circuit component values

Figure 4.8: Schematic of grounding diagram

4.3.1.3 Calibration

4.3.1.3.1 Calibrating power supply output

4.3.1.3.2 Calibrating cap bank output

4.3.2 Probe Circuit

4.3.2.1 Single-ended measurement

4.3.2.1.1 TREX cells

4.3.2.1.2 Why was this bad

4.3.2.1.3 How long did we spend on this :(

4.3.2.2 Differential measurement

4.3.2.2.1 Parallel circuitry

4.3.2.3 Figuring out grounds

4.3.2.3.1 Types of grounds on the ACQ

4.3.2.3.2 Routing grounds

Figure 4.9: Image of swapping box

Figure 4.10: Diagram of swapping

Figure 4.11: Time frame for core movie of IV with separation between relay

4.3.2.4 Swapping current paths

4.3.2.4.1 Why do we want to do this

4.3.2.4.1.1 Takes care of small differences in resistors

4.3.2.4.2 Circuitry to do this

4.3.2.4.2.1 Only swap core stuff

4.3.2.4.2.2 Many relays

4.3.2.4.2.3 Remote control

4.3.2.4.3 How does this change results

4.3.2.5 Circuit Solution

4.3.2.5.1 Solution

4.3.2.5.2 Assumptions

4.3.2.6 Calibration

4.3.2.6.1 Calibrating in isat

Figure 4.12: Circuit diagram of probe circuitry

Figure 4.13: Image of probe circuitry

Figure 4.14: Probe signals showing coupling

4.3.2.6.2 Calibrating on the exponential

4.3.2.6.3 Rotation calibration

4.3.2.7 The path to ground

4.3.2.7.1 Why do we have to care

4.3.2.7.2 How do we set it

4.3.2.8 Capacitive coupling

4.3.2.8.1 Where does it come from

4.3.2.8.2 Where do we see it

4.3.2.8.3 Magnitude of effect

4.3.2.8.4 Removing effect

Figure 4.15: Figure of theoretical plasma distributions and JV curves

4.4 Plasma Theory

4.4.1 A full model

4.4.1.1 Plasma Distribution

4.4.1.2 Ion particle flux

4.4.1.2.1 Mach effects

4.4.1.3 Electron particle flux

4.4.1.3.1 Minimum velocity

4.4.1.3.2 Area

4.4.1.3.3 Particle Flux

4.4.1.4 Theoretical JV curves

4.4.1.4.1 Reading radar plots

4.4.1.4.1.1 Radar axis

4.4.1.4.1.2 Solutions

4.4.2 Individual tip fitting

4.4.2.1 Why do this instead

4.4.2.2 Methods

4.4.2.2.1 Aforementioned methods

Figure 4.16: Simple drawing of what might happen to ions

4.4.2.2.2 Exponential

4.4.2.2.2.1 Ions don't hit core

4.4.2.2.2.2 Take care of this by allowing free parameters

4.4.2.2.2.3 Converting from plasma parameters to exponential

4.4.2.2.2.4 Converting back

4.4.2.2.2.5 Taking care of errors

4.4.2.3 Extracting plasma parameters

4.4.2.3.1 Spherical harmonics

4.4.2.3.2 Error bars

4.4.2.3.2.1 Monte-carlo

4.4.2.3.2.2 Other options

4.4.2.3.2.3 Why can we ignore non-bi-maxwellian features

4.5 Curve Fitting

We will use ODR like mentioned in Te section.

Figure 4.17: Comparison of alignment using $B_z = 0$ or $\mathbf{b} \cdot \mathbf{d}$ cross correlation

4.5.1 Combining multiple shots

4.5.1.1 Timing alignment

4.5.1.1.1 $B_z = 0$ alignment

4.5.1.1.1.1 Aligns where we expect Anisotropy

4.5.1.1.1.2 Depends heavily on prior $\mathbf{b} \cdot \mathbf{d}$ data

4.5.1.1.2 $\mathbf{b} \cdot \mathbf{d}$ cross correlation

4.5.1.1.2.1 Still requires smoothing to get rid of initial oscillation impact

4.5.1.1.3 Result comparison

4.5.1.2 Averaging of core data

4.5.1.2.1 Weighted average of relay on/off

4.5.2 Fitting shell

4.5.2.1 Use all shell data

4.5.2.2 Fit using BRL

4.5.2.3 Fits pretty good

4.5.3 Fitting core

4.5.3.1 Fitting using full routine

4.5.3.2 Fitting tips independently

4.6 Results

Figure out what my results are once I have some reasonable results

4.6.1 Drive cylinder

4.6.1.1 What did we try

4.6.1.2 Why didn't we see pressure anisotropy

4.6.1.2.1 In the bdot data

4.6.1.2.2 In the PA data

4.6.2 3 coil

4.6.2.1 How much anisotropy should we expect

4.6.2.2 What setups did we try

4.6.2.3 Old results of just comparing radar plots

4.6.2.4 New results using independent fitting

5 CONCLUSION

5.1 Summary of thesis

- 5.1.1 Using pressure waves to measure density
- 5.1.2 Upgrading langmuir probe analysis and new possibilities
- 5.1.3 Using directional langmuir probes for measuring pressure anisotropy

5.2 Future work

- 5.2.1 Pressure wave
 - 5.2.1.1 Aforementioned future work
 - 5.2.1.2 Extracting cross-field diffusion
- 5.2.2 Langmuir Probe
 - 5.2.2.1 Prepared to analyze lots of cool data
 - 5.2.2.2 Extract out electric field
- 5.2.3 PA Probe
 - 5.2.3.1 Try more configurations
 - 5.2.3.2 Map out full pressure anisotropy during shot

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